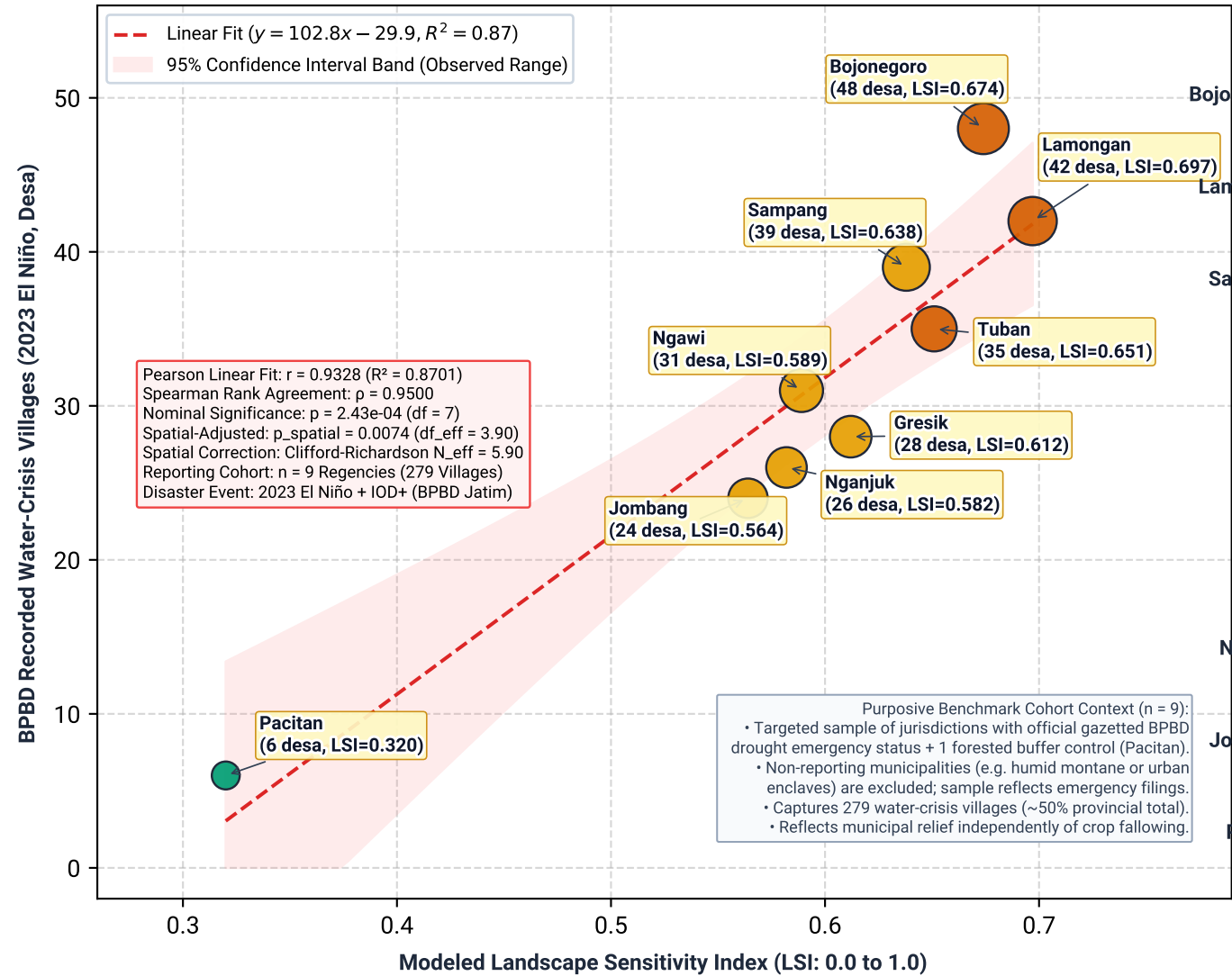


Empirical Disaster Ground-Truth Validation: Modeled Sensitivity vs BPBD 2023 Drought Records

Statistical Concordance & Water-Crisis Village Incidents in Purposive Benchmark Reporting Cohort (n = 9) | Official BPBD Jawa Timur PUSDALOPS Data (2023 Compound El Niño)

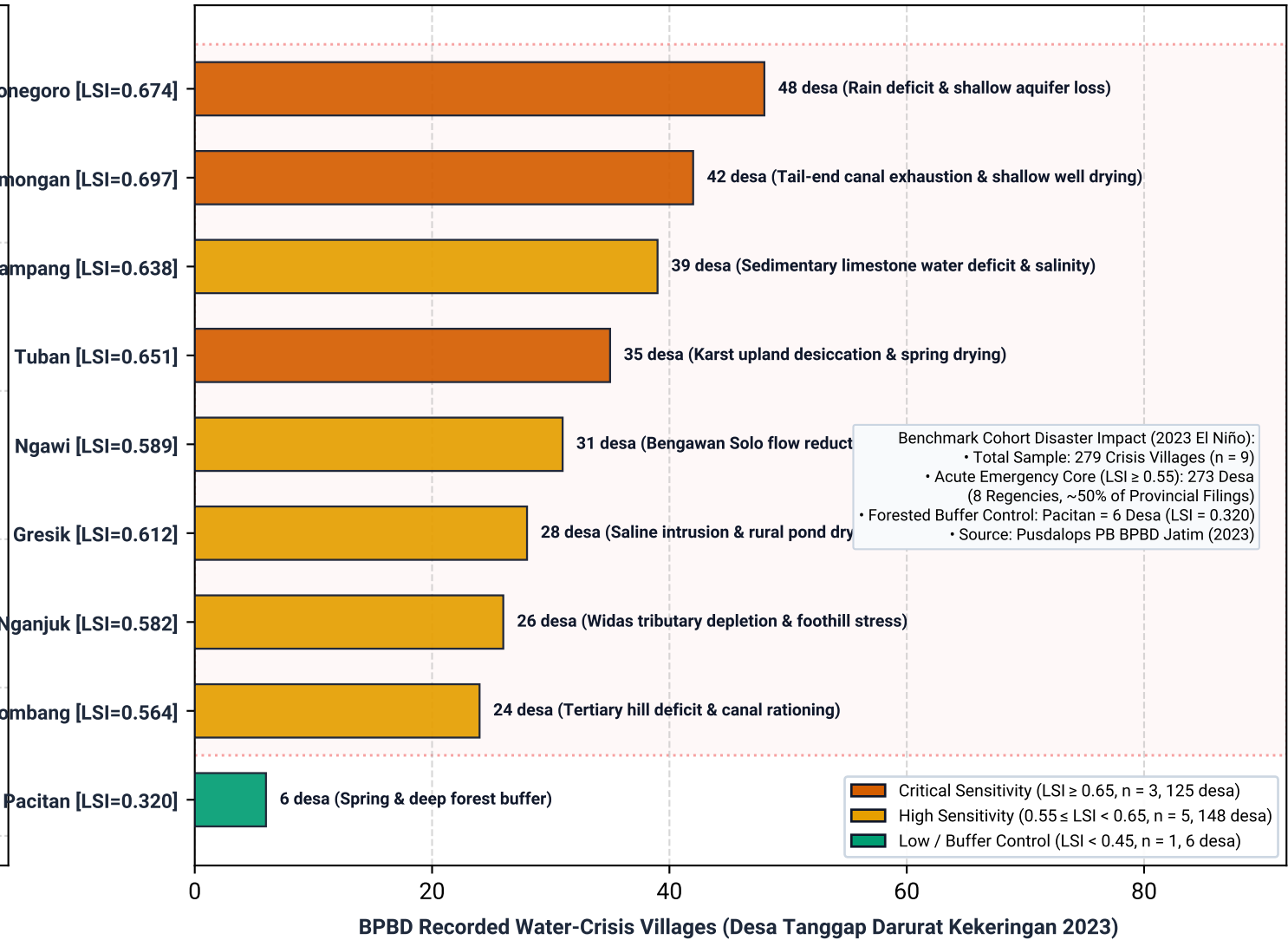
(a) Statistical Concordance & Incident Correlation

Modeled Landscape Sensitivity (LSI) vs BPBD Water-Crisis Villages | Pearson $r = 0.933$ ($R^2 = 0.870$)



(b) Empirical Disaster Incident Profile & Local Stress Drivers

BPBD Water-Crisis Villages & Emergency Drought Declarations (2023 El Niño) | Benchmark $n = 9$



- Strong Sample Association: Modeled LSI accounts for 87.0% of sample variance in recorded crisis villages ($r = 0.9328$, $R^2 = 0.8701$, $\rho = 0.9500$).
- Spatially Adjusted Significance: Association remains significant after Clifford-Richardson spatial correction ($p_{\text{spatial}} = 0.0074$, $N_{\text{eff}} = 5.90$, $df_{\text{eff}} = 3.90$).
- Acute Concentration: 8 sensitive regencies ($LSI \geq 0.55$) encompass 273 crisis villages, capturing ~50% of provincial emergency filings.

- Purposive Sampling Provenance: Benchmark cohort reflects jurisdictions with gazetted BPBD emergency filings during 2023; non-reporting areas excluded.
- Operational Relevance & Confounders: Measures municipal water relief independently of crop fallowing, though administrative capacity & infrastructure remain unmodeled.
- Empirical Support: High statistical concordance supports the utility of satellite LSI for regional drought vulnerability screening, subject to broader testing.